Designing a College Department Website using HTML

Web Programming — Assignment Report

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1. Introduction

This report explains how a simple multi-section website for the **Department of Computer Science and Engineering** was created using only HTML (with a small internal style sheet for colours). The page contains six sections — About, Programmes, Faculty, Facilities, Enquiry and Contact — which are reached through a navigation bar at the top. The page was written in VS Code and tested in Google Chrome.

2. HTML Document Structure

Every HTML page follows a fixed skeleton. The skeleton used for the department page is shown below.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Department of Computer Science and
Engineering</title>
</head>
<body>
  ... visible content of the page ...
</body>
</html>
```

- `<!DOCTYPE html>` – tells the browser that the document uses HTML5, so it renders in standards mode.
- `<html>` – the root element that encloses the whole page; `lang="en"` sets the language.
- `<head>` – holds information about the page (charset, title, styles) which is not shown in the body.
- `<title>` – text shown on the browser tab and used by search engines and bookmarks.
- `<body>` – contains everything the user actually sees: headings, text, images, tables and forms.

3. Headings, Paragraphs and Text Formatting

HTML provides six heading levels, `<h1>` (most important) to `<h6>`. The department name is an `<h1>`, each section title is an `<h2>` and sub-titles are `<h3>`. Text is written inside `<p>` paragraphs. Formatting tags used are `<strong>`/`<b>` (bold), `<em>` (italic emphasis), `<mark>` (highlight) and `<br>` (line break).

```
<h2>About the Department</h2>
<p>The Department of <strong>Computer Science and Engineering</strong> was started in 2001 ...
   Today it is one of the <em>largest departments</em> of the college ...</p>
<p>Our mission is to produce engineers who can <mark>solve real-world problems</mark> using
computing.</p>
```

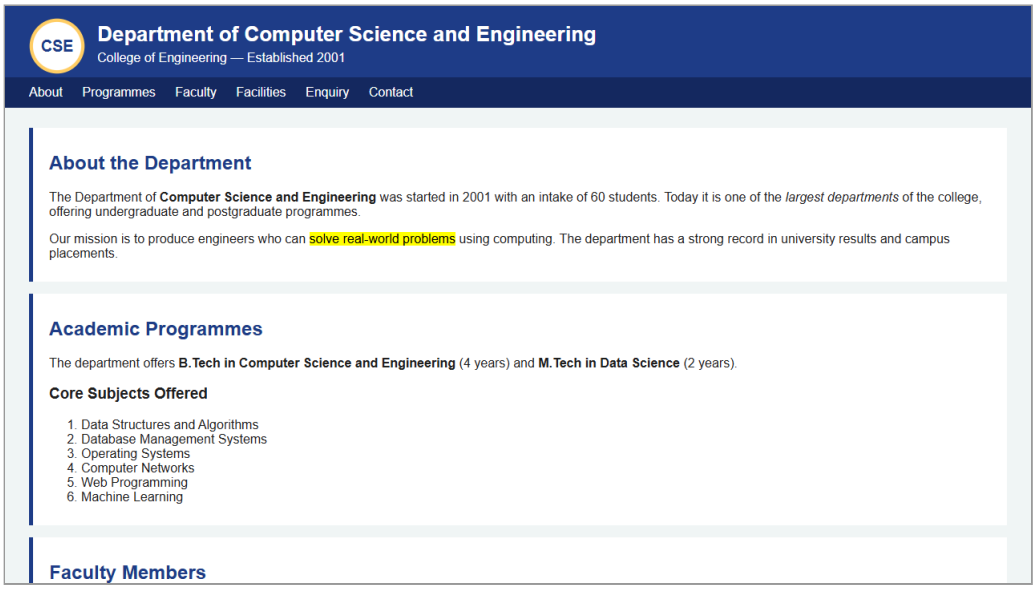


Figure 1: Header, navigation bar and the About section (headings, paragraphs, bold, italic and highlight)

4. Lists

Lists group related items. An **ordered list** `<ol>` numbers its items automatically, while an **unordered list** `<ul>` uses bullets. Each item is written inside `<li>`. The subjects are shown as an ordered list and the facilities as an unordered list.

```
<h3>Core Subjects Offered</h3>
<ol>
  <li>Data Structures and Algorithms</li>
  <li>Database Management Systems</li>
  <li>Operating Systems</li>
  <li>Computer Networks</li>
  <li>Web Programming</li>
  <li>Machine Learning</li>
</ol>
```

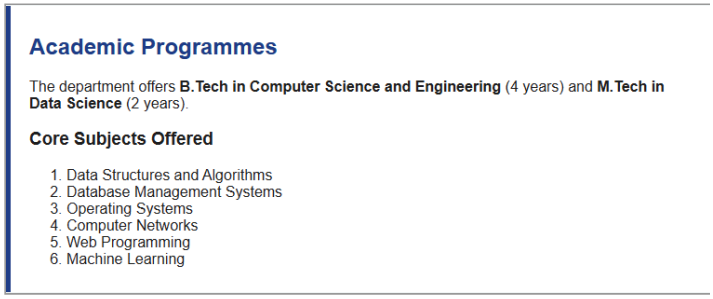


Figure 2: Ordered list of subjects

5. Hyperlinks

The anchor tag `<a>` creates links. The `href` attribute gives the destination. Three kinds of links are used: **internal links** (`href="#faculty"`) in the navigation bar that jump to a section with that `id`, **external links** to the college website and other sites (opened in a new tab with `target="_blank"`), and an **e-mail link** using `mailto:`.

```
<nav>
  <a href="#about">About</a>  <a href="#faculty">Faculty</a>  <a href="#enquiry">Enquiry</a> ...
</nav>
<a href="https://www.example.edu" target="_blank">College Website</a> |
<a href="https://ktu.edu.in" target="_blank">KTU</a> |
<a href="https://nptel.ac.in" target="_blank">NPTEL</a>
Email: <a href="mailto:cse@example.edu">cse@example.edu</a>
```



Figure 3: Contact section with e-mail and external links

6. Images

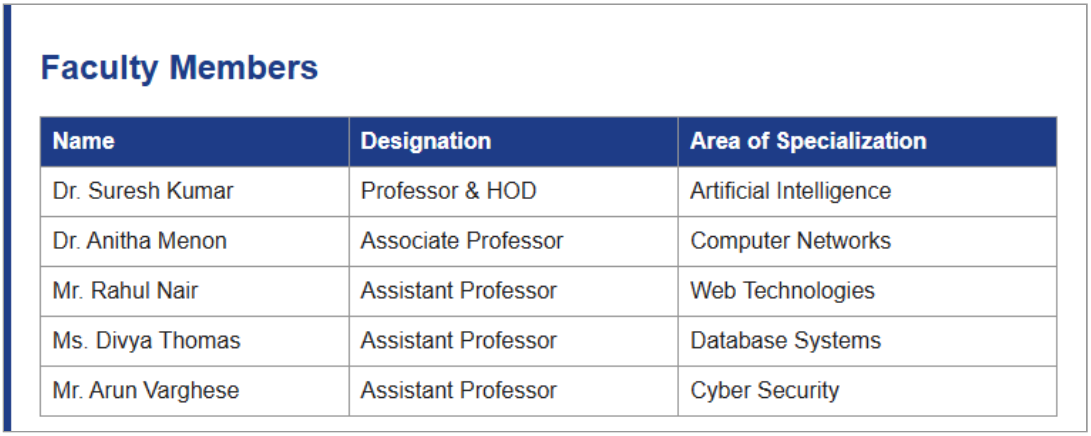
Images are inserted with the empty tag `<img>`. The `src` attribute gives the path of the image file, `alt` gives alternate text that is read by screen readers and shown if the image fails to load, and `width`/`height` set its size in pixels. The department logo (`images/logo.svg`) is placed in the header, as seen in Figure 1.

```
<img src="images/logo.svg" alt="CSE Department Logo" width="70" height="70">
```

7. Tables

A table is created with `<table>`. Each row is a `<tr>`; header cells use `<th>` (bold and centred by default) and data cells use `<td>`. Borders were added using CSS `border` and `border-collapse`.

```
<table>
  <tr> <th>Name</th> <th>Designation</th> <th>Area of Specialization</th> </tr>
  <tr> <td>Dr. Suresh Kumar</td> <td>Professor &amp;amp; HOD</td> <td>Artificial Intelligence</td> </tr>
  <tr> <td>Dr. Anitha Menon</td> <td>Associate Professor</td> <td>Computer Networks</td> </tr>
  ... three more rows ...
</table>
```



Faculty Members

Name	Designation	Area of Specialization
Dr. Suresh Kumar	Professor & HOD	Artificial Intelligence
Dr. Anitha Menon	Associate Professor	Computer Networks
Mr. Rahul Nair	Assistant Professor	Web Technologies
Ms. Divya Thomas	Assistant Professor	Database Systems
Mr. Arun Varghese	Assistant Professor	Cyber Security

Figure 4: Faculty table with five members

8. Forms

Forms collect input from the user. The `<form>` tag has an `action` (where data is sent) and a `method` (get/post). `<label>` describes each field and is linked to it by the `for` attribute. Different `type` values give built-in validation: `text`, `email` (checks for @) and `tel`. A drop-down list is made with `<select>` and `<option>`, and `type="submit"` creates the submit button. The `required` attribute stops the form from being sent when a field is empty.

```
<form action="#" method="post">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name"
required><br>
  <label for="email">Email:</label>
  <input type="email" id="email"
name="email" required><br>
  <label for="phone">Phone Number:</label>
  <input type="tel" id="phone" name="phone"
pattern="[0-9]{10}"><br>
  <label for="course">Course Preference:
</label>
  <select id="course" name="course">
    <option>B.Tech CSE</option>
    <option>M.Tech Data Science</option>
  </select><br>
  <input type="submit" value="Submit
Enquiry">
</form>
```

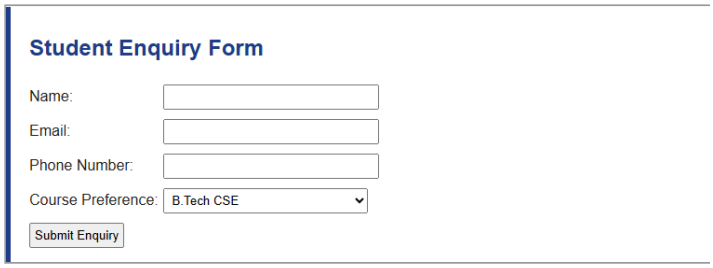


Figure 5: Student enquiry form

9. Semantic HTML

Semantic elements describe the *meaning* of their content instead of just its appearance. They make the code easier to read and help search engines and screen readers understand the page. The following were used:

Element	Purpose	Use in this website
`<header>`	Introductory content of a page	Logo and department name
`<nav>`	Block of navigation links	Links to all six sections
`<main>`	Main, unique content (one per page)	Wraps all the sections
`<section>`	Thematic group of content with a heading	About, Programmes, Faculty, etc.
`<footer>`	Closing information of a page	Copyright and “Back to top” link

```
<body>
  <header> ... </header>
  <nav> ... </nav>
  <main>
    <section id="about"> ... </section>
    <section id="faculty"> ... </section>
  </main>
  <footer> ... </footer>
</body>
```

10. Conclusion

Using basic HTML tags a complete, readable department website was built. Structure tags gave the page its skeleton, semantic tags organised it into meaningful parts, and lists, tables, links, images and forms presented the department’s information. The source code has been uploaded to GitHub.

References

1. MDN Web Docs – HTML: HyperText Markup Language, https://developer.mozilla.org/en-US/docs/Web/HTML
2. W3Schools – HTML Tutorial, https://www.w3schools.com/html/
3. WHATWG – HTML Living Standard, https://html.spec.whatwg.org/
4. Jon Duckett, *HTML and CSS: Design and Build Websites*, Wiley, 2011.